

Case 16: No-spread limiting behavior

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE16_NSP. This unit test verifies the zero-normal-velocity limit of the ELMFIRE level-set pathway. A zero ADJ raster suppresses the base spread rate while wind, slope, spotting, crown fire, and suppression are disabled. The case exercises input multiplication, propagation, and phi/vs output and can expose ignored adjustment rasters, nonzero numerical forcing, or unintended growth. It is implementation verification, not physical validation. The question is whether the final rate remains below $10^{-6} \text{ m min}^{-1}$ and the ignited footprint remains confined to the initialization band.

2 Mathematical formulation and numerical implementation

The propagation equation is $\partial_t \phi + V_n |\nabla \phi| = 0$. The case sets the multiplicative adjustment to zero, so $V_n = 0$ and therefore $\partial_t \phi = 0$. ELMFIRE still initializes a grid-resolved ignition neighborhood, so exact zero burned area is not expected; the invariant is no subsequent outward propagation.

3 Derivation of expected behavior

Integrating $\partial_t \phi = 0$ gives $\phi(\mathbf{x}, t) = \phi(\mathbf{x}, 0)$. Thus the output rate is zero and any negative- ϕ area must be limited to the rasterized initialization band. A 2.5-cell equivalent-radius allowance bounds that band without treating subcell front placement as spread.

The generated pre-ignition ϕ raster contains no negative cells and the adjustment raster is identically zero. Thus

$$V_n = R_0(ADJ) = R_0(0) = 0, \quad \phi(\mathbf{x}, 10 \text{ s}) = \phi(\mathbf{x}, 0), \quad R_{\text{ref}} = 0 \text{ m min}^{-1}.$$

The rate metric is therefore the absolute quantity $e_R = \max_i |R_i|$, with the substituted test $e_R \leq 10^{-6} \text{ m min}^{-1}$; a relative error is undefined for the zero reference. ELMFIRE may rasterize its point ignition into a small initialization footprint, so the spatial metric is

$$r_A = \sqrt{\frac{N_b (1 \text{ m})^2}{\pi}},$$

where N_b counts final cells with $\phi < 0$. Its no-propagation reference is zero and the explicit grid allowance is $r_A \leq 2.5(1 \text{ m}) = 2.5 \text{ m}$.

4 Verification metrics and acceptance criteria

The maximum absolute finite final vs value tests the zero-velocity invariant directly and must not exceed $10^{-6} \text{ m min}^{-1}$. The equivalent burned radius $r_A = \sqrt{N_b \Delta x^2 / \pi}$ must be no more than $2.5 \Delta x$;

this whole-domain area measure detects growth in any direction and is more robust than one transect. Nodata is excluded, and the last matching final `phi` and `vs` rasters are required. Both conditions must pass; missing output is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The generated EPSG:32610 domain is 200 m square on a 200-by-200 grid. Fuel, moisture, and terrain are uniform; wind and slope are zero; `ADJ=0`; the ignition is centered; and duration is 10 s. The predicted `vs` field is zero and the final ignition footprint does not expand beyond its grid initialization band.

Parameter	Configured value	Units	Role
Domain/grid	200 by 200 cells; 1	m/cell	Resolves the stationary ignition
Initial front	centered rasterized band	—	Nonzero initialized footprint
Adjustment	<code>ADJ=0</code>	—	Forces zero normal speed
Time control	step 1; stop 10	s	Tests the zero-speed invariant
Disabled forcing	wind, slope, crown, spotting, suppression	—	Excludes alternate growth pathways

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

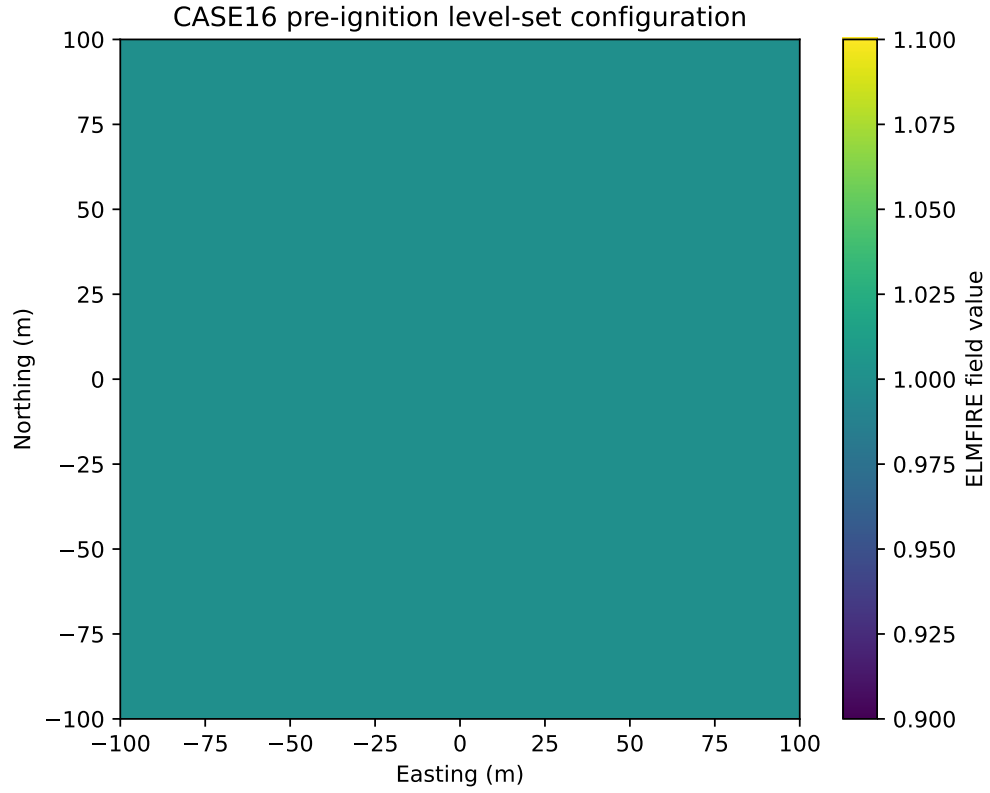


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-local ELMFIRE rasters are read. Current status is **PASS**; 2 of 2 components pass.

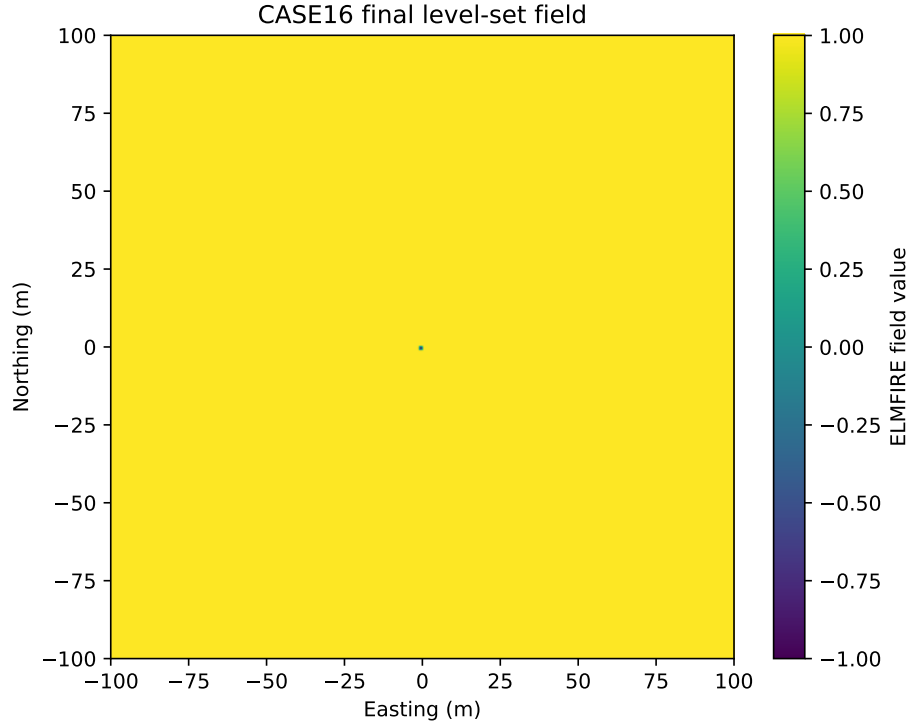


Figure 2: Final whole-domain signed level-set field for the completed zero-spread variant at the final output time. Values are dimensionless; the generated table supplies the rate and confinement limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
maximum absolute spread rate	0 ($\leq 1\text{e-}06$)	0 m/min	PASS
equivalent burned radius	0 (≤ 2.5 cells (2.5 m))	0.56419 m	PASS

The authoritative overall decision is **PASS**. The generated table is sourced directly from the case-local metrics JSON and is not recomputed in this report.

8 Technical interpretation

A nonzero rate primarily indicates that the zero adjustment was ignored or transformed incorrectly. A confined rate with excessive burned area indicates initialization or level-set evolution behavior. Unavailable output cannot establish the invariant.

9 Scope and limitations

This case verifies a deterministic limiting state and raster adjustment coupling. It does not establish accuracy for positive rates, physical extinction, or any coupled fire process.